APT-BENCH: BENCHMARKING CROSS-PATIENT CELL TYPING FROM SINGLE-CELL APTAMER PROFILES

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ABSTRACT

Which cell identities can aptamer profiles recover in new patients, and what do they add beyond RNA? APT-Bench specifies this comparison using 361,792 paired cells from 40 participants, 293 aptamer measurements, five lineages, 27 RNA-derived subtypes, and fixed patient-disjoint folds. Two findings emerge. First, the full-training-budget MLP achieves fine subject-balanced Macro-F1 of 0.141, rising to 0.385 with true-lineage decoding, compared with 0.180 for a training-only conditional-proportion prior. This comparison shows additional discrimination relative to the tested training-only lineage priors; matched-budget APT/RNA comparisons reveal substantial lineage heterogeneity. Second, adding paired APT improves MLP fine Macro-F1 by 0.0180 and 0.0165 over RNA references using 2,000 and 5,000 training-selected genes, respectively. Yet the overall advantage over APT shuffled within patient and true lineage remains statistically unresolved. Thus an increment over these RNA references is established without establishing finer-grained pairing specificity. Classical and neural APT-only baselines anchor a reusable new-patient comparison, separating recoverable identity from claims of an intrinsic assay limit or a biological mechanism.

1 INTRODUCTION

Single-cell aptamer profiling links cell-surface binding measurements to transcriptomic annotations. Apt-seq demonstrated this principle in cell-line mixtures (Delley et al., 2018), and subsequent Aptomics work extended molecular profiling to clinical specimens (Wu et al., 2025). These capabilities motivate a distinct evaluation question: in patients absent from training, which annotated identities can APT recover, and does paired APT improve on a specified RNA-only reference?

Two comparisons are needed to answer this question. First, low fine-subtype performance does not establish a lack of APT information: parent-lineage errors, candidate-set size, annotation ambiguity and the fitted model all affect reconstruction. True-lineage decoding must therefore be compared with a training-only prior given the same privileged lineage. Second, an RNA+APT gain can reflect patient structure or coarse-lineage correspondence rather than finer cell pairing. Shuffling APT within patient and then within patient and lineage provides progressively more specific controls.

APT-Bench brings these comparisons into one clinical PBMC evaluation setting: 361,792 paired cells from 40 participants, 293 APT measurements, and a fixed five-lineage/27-subtype ontology. Five patient-disjoint folds and subject-balanced pooled Macro-F1 define a common new-patient target. Classical and neural APT-only models establish reference performance; Soft-cascade is one hierarchy-aware reference rather than a separate algorithmic contribution. Our contributions are:

- **An APT cell-typing evaluation resource.** A fixed input panel, operational hierarchy, patient folds, explicit scoring units and reference predictions support comparisons on the same target. Complete data access and portable reproduction remain release requirements.
- **Within-lineage discrimination beyond a candidate-set prior.** The full-budget MLP reaches oracle fine F1 of 0.385 versus 0.180 for a conditional-proportion baseline. Matched-budget APT/RNA results reveal heterogeneous recoverability across lineages, rather than a uniform assay-resolution deficit inferred from coarse/fine scores.
- **An RNA-reference increment with a bounded interpretation.** Paired APT improves MLP fine F1 by 0.0180 and 0.0165 at two RNA widths. Its overall patient-by-lineage-

shuffle contrast remains unresolved: improvement over these RNA references does not yet establish finer-grained pairing specificity.

The resulting findings concern recoverability under a specified cohort, taxonomy and training recipe. RNA-derived annotations are not independent biological truth, and an increment over a restricted RNA representation is not evidence that RNA intrinsically lacks that information. Patient-disjoint testing supplies the evaluation foundation, not evidence of blood-based diagnostic validity.

2 Related Work

From aptamer profiling to a typing evaluation. Apt-seq jointly measured aptamer binding and RNA and used Ramos/3T3 mixtures to demonstrate cell discrimination (Delley et al., 2018). Aptomics subsequently profiled proteins, glycans and RNA in cell lines and clinical specimens, including tumor heterogeneity and T-cell differentiation (Wu et al., 2025). SPARK-seq integrates CRISPR perturbations with aptamer and RNA sequencing to identify binding targets and kinetics (Luo et al., 2026). These studies establish measurement and molecular profiling capabilities. The complementary question addressed here is how a fixed clinical APT panel reconstructs an annotated hierarchy in unseen patients, separating a restricted candidate set from within-lineage discrimination and an RNA-reference increment from conditional pairing specificity. We evaluate these distinctions jointly without claiming priority for paired profiling or all aptamer-based cell classification.

Multimodal annotation and reference models. Integrated RNA/protein analysis makes complementary measurements useful for cell-state characterization (Hao et al., 2021); hierarchical annotation methods exploit relationships among labels (Jia et al., 2025; Cultrera di Montesano et al., 2026). Large transcriptomic representations offer another axis of comparison, but their value depends on the task and baseline (Theodoris et al., 2023; Cui et al., 2024; Kedzierska et al., 2025). Here, input-compatible classical and neural models anchor APT-only performance, while a separate, budget-matched RNA experiment addresses complementarity. Its oracle and prior controls are diagnostic comparisons, not deployable models or independent validation of RNA-derived labels.

Reusable single-cell evaluation. Open Problems separates tasks, datasets and metrics (Luecken et al., 2025); multimodal benchmarks compare input configurations (Liu et al., 2025). APT-Bench specializes this approach to a fixed aptamer panel and clinical hierarchy. Its resource value is the common new-patient target and explicit controls for interpreting reconstruction, not a new principle of subject splitting. This scope also motivates strong tabular references rather than relying solely on architectural complexity (Grinsztajn et al., 2022; Erickson et al., 2026).

3 Benchmark Setting

Blood-derived profiles
293 APT measurements per cell
Paired RNA supplies reference annotations

⟶

Split by patient
Five fixed outer folds
No patient shared between development and test

⟶

Evaluate cell typing
5 lineages / 27 subtypes
Equal patient weight; frozen APT-only references

Figure 1: The primary APT-Bench contract. Patient-disjoint outer folds evaluate cell typing in new individuals; subject-normalized confusion pooling assigns equal total scoring weight to each patient.

3.1 Blood Profiles and Reference Labels

APT-Bench is derived from peripheral-blood specimens from 40 participants in five cancer groups (CRC, GC, HCC, PC, and BTC) and healthy controls, with 6–8 participants per group. The paired assay measures transcriptomic state and 293 target-specific nucleic-acid aptamers on the same cells. Transcriptomic counts were processed with Seurat (Hao et al., 2021); predicted doublets and low-quality cells were removed, and clusters were manually annotated using canonical markers from prior PBMC and pan-cancer studies (McGinnis et al., 2019; Zhang et al., 2024; Yang et al., 2024). These annotations define five lineages and 27 subtypes. They are operational reference labels, not independently measured biological truth.

The linked post-QC cohort contains 374,854 cells. Excluding the uncertain *Unknown* subtype leaves 361,792 benchmark cells. Each subtype has one fixed parent lineage. Disease is a six-class patient attribute, not the root of the cell hierarchy: any given lineage can occur in multiple disease groups. We evaluate the lineage-to-subtype hierarchy without imposing a disease-to-lineage-to-subtype label tree. The construction checklist and unresolved acquisition and annotation fields are recorded in Appendix B.1.

3.2 Patient-Disjoint Cell-Typing Contract

The primary task is cell typing in new patients. **Cell identity** predicts lineage and subtype using APT alone at inference. Training supervision is recipe-specific: only Soft-cascade additionally uses development-patient disease labels for contrastive pairing. Open-set subtype discovery is outside the contract; all primary cell tasks use the fixed five-lineage/27-subtype ontology.

Five frozen disease-stratified folds each hold out eight patients. All cells from a patient remain together; preprocessing, model selection, and fitting are restricted to development patients. We pool predictions across the five out-of-fold (OOF) partitions. The release interface records cell and patient IDs, ontology, folds, predictions, and metrics (Appendix B.2). Full data access and a portable end-to-end evaluator remain release requirements.

3.3 Metrics and Statistical Units

Coarse and fine subject-balanced pooled Macro-F1 (SB-Macro-F1) are the primary cell metrics. For patient p, let $C^{(p)}$ be the confusion matrix over a fixed K-class ontology and n_p its number of evaluated cells. Define

$$\overline{C} = \frac{1}{P}\sum_{p=1}^{P}\frac{C^{(p)}}{n_p}, \qquad \text{SB-MacroF1} = \frac{1}{K}\sum_{k=1}^{K}\frac{2\overline{C}_{kk}}{\sum_j \overline{C}_{kj} + \sum_i \overline{C}_{ik}}. \tag{1}$$

A zero denominator contributes zero. Each patient's entire matrix, not each class row, is normalized to unit mass. This differs from cell-weighted pooled Macro-F1 and from mean within-subject Macro-F1. We retrospectively re-score the frozen OOF predictions under this primary reporting convention; checkpoints are not reselected.

Main-table intervals use 2,000 resamples and explicitly label their target: P denotes patient uncertainty conditional on frozen fits, and SP additionally resamples the three MLP training seeds. Each SP replicate averages three seed-specific scores, using the same patient draw across them. Both are pointwise and conditional on the fixed fold design; neither repeats model selection. The comparison is descriptive, and an individual-score interval is not a pairwise superiority test. Full resampling order is given in Appendix B.3.

The paired-RNA probe also uses subject-balanced pooled Macro-F1, but has a different training budget and preprocessing protocol; its scores must not be interpreted as additional architecture rows in the main table. The experimental protocol is in Appendix B.3; scoring-unit comparisons are in Appendix B.7.

4 Benchmark Reference Models

APT-only comparisons. LR and XGBoost provide linear and nonlinear tabular references. Flat and HCE Transformers and a Soft-cascade Transformer provide backbone-matched neural recipes. All use APT at inference and the same outer patient partitions. Training and validation follow each recipe's documented protocol; identical outer folds do not imply identical training-set sizes or fully matched tuning. Linear SVM, Random Forest, and reference correlation are additional appendix references. A plain MLP uses all development cells in each outer fold, with three fixed training seeds, inner-patient selection, and outer refitting. Its 256/128-unit ReLU network uses dropout 0.1, AdamW, and training-fold standardization; the inner fold selects weight decay and epoch (at most

50). This completes a full-cell-budget reference, not an objective- or optimizer-matched ablation of the Transformers.

Hierarchy-aware reference. Soft-cascade Transformer embeds aptamer identities and values and combines a global subtype head with lineage-specific experts weighted by coarse probabilities. Appendix A specifies the routing equation and training objective. The independent coarse and fine heads are evaluated without constrained decoding. Flat removes the cascade; HCE instead uses a two-level hierarchical loss adaptation (Cultrera di Montesano et al., 2026). Only Soft-cascade uses cross-disease supervised contrastive pairing, so their objectives and auxiliary disease-label usage differ. These are reference-recipe comparisons, not component ablations. No routing or contrastive contribution is claimed in isolation. Appendix A gives the full architecture, losses, hyperparameters, and component matrix.

Input compatibility. Gene-based pretrained models such as scGPT and Geneformer (Cui et al., 2024; Theodoris et al., 2023) do not directly accept anonymous aptamer feature IDs without a verified mapping or a separately trained adapter. Supplying paired RNA would change the input contract. This distinguishes an undefined direct APT input from an adaptable but unevaluated model; it does not justify claiming superiority over transcriptomic annotation methods.

Matched-input diagnostics. A separate matched-input probe varies APT and RNA while holding its training budget and patient folds fixed. Training-only subtype priors and true-lineage masking provide candidate-set controls. APT shuffling within patient, or within patient and true lineage, is a privileged diagnostic rather than a deployable model input (Appendix B.3). RNA also supplies the reference annotations, so this is an internal reconstruction comparison, not independent label validation. HVG selection and scaling are fitted separately within each inner and outer training partition. Table 4 gives the input transforms, training weights, optimization settings and selection grids.

5 Results

5.1 Reference Performance in New Patients

Table 1 reports APT-only typing on the same 40 held-out patients. LR and XGBoost reach coarse/fine SB-Macro-F1 of 0.298/0.126 and 0.325/0.137; the full-budget MLP reaches 0.334/0.141. The Transformer references achieve fine scores of 0.139–0.145 without an established Soft-cascade advantage. These are reference-recipe comparisons, not objective-matched ablations. The interval labels distinguish conditional patient uncertainty from the MLP's joint seed/patient uncertainty.

Model	CI	Coarse SB-Macro-F1 ↑	Fine SB-Macro-F1 ↑
Logistic Regression	P	0.298 [0.281, 0.315]	0.126 [0.107, 0.139]
XGBoost	P	0.325 [0.302, 0.345]	0.137 [0.120, 0.149]
Plain MLP (3 seeds)	SP	0.334 [0.311, 0.355]	0.141 [0.125, 0.155]
Flat Transformer	P	0.326 [0.300, 0.348]	0.139 [0.114, 0.155]
HCE Transformer	P	0.335 [0.310, 0.356]	0.143 [0.121, 0.160]
Soft-cascade Transformer	P	0.332 [0.306, 0.354]	0.145 [0.122, 0.160]

Table 1: Primary APT-only cell-typing comparison: five patient-disjoint folds, 40 patients, 361,792 cells. All inputs are APT. Entries are subject-balanced pooled Macro-F1 with pointwise 95% intervals (2,000 resamples). CI type P conditions on frozen fits and resamples patients; SP resamples both training seeds and patients for the MLP three-seed mean. These uncertainty targets differ; interval widths are not directly comparable evidence of model stability. Comparisons are descriptive: training recipes and Transformer auxiliary supervision differ (Table 3).

Five-lineage and 27-subtype tasks differ in cardinality, prevalence and label boundaries. Their raw F1 difference alone does not identify an APT resolution limit. The following prior-controlled comparisons instead ask what remains recoverable within a known lineage. Full subtype reporting and the supporting split-protocol comparison are retained in Appendices B.6 and B.7.

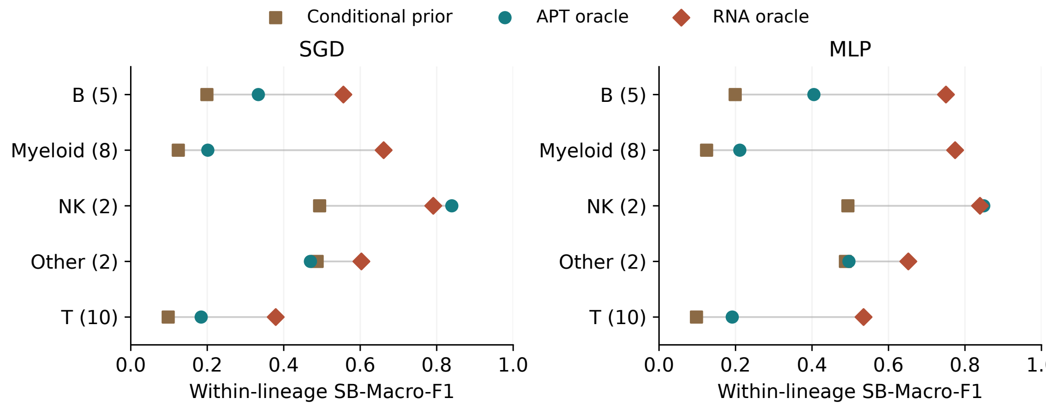

Figure 2: Matched-budget within-lineage reconstruction with 2,000 training-selected RNA HVGs. APT oracle, RNA oracle and conditional priors use the same capped training subsets, patient folds and true-lineage information. Points are means of three seed-specific scores, not F1 of seed-pooled predictions; lines only connect methods within a lineage. Each patient's matrix is normalized within the displayed lineage before pooling over eligible patients and scoring its fixed children (38 patients for Other; 40 elsewhere). Consequently, these lineage scores cannot be averaged to reconstruct the All row of Table 5 from this same probe. Row labels give the number of child subtypes. These are descriptive point estimates, not superiority tests. Expected-prior F1 is computed from an expected confusion matrix.

5.2 Within-Lineage Identity Is Recoverable, but Heterogeneous

True-lineage decoding masks each frozen fine score vector to the correct parent's children. Both the model and the training-only conditional priors receive this privileged label. For the full-budget MLP, ordinary fine F1 is 0.141 and oracle F1 is 0.385, versus 0.148 for conditional majority decoding and 0.180 for the conditional-proportion expected-confusion baseline. The oracle-minus-expected-prior difference is $+0.205$ (joint 95% CI $[0.174, 0.217]$). Thus APT supports discrimination beyond these candidate-set controls under this recipe; the comparison does not measure all information in the assay.

The matched-input probe holds budget and recipe fixed across modalities (Figure 2). MLP ordinary/oracle APT scores are 0.093/0.309, RNA oracle is 0.672, and the expected prior is 0.182. The APT oracle-minus-prior difference is $+0.127$ ($[0.101, 0.139]$). Relative to each lineage's own conditional prior and RNA reference, APT reconstruction differs across lineages. Their absolute F1 values are not a common biological-resolution scale: child-class counts and prevalence also differ. For example, APT/RNA oracle scores are 0.849/0.839 for NK but 0.191/0.535 for T cells. Knowledge of the parent therefore does not uniformly resolve the RNA reconstruction gap. All lineages and both model families are shown, without asserting an intrinsic modality ranking.

Budget and recipe matter. The full-budget MLP uses all development cells, a 256/128-unit network and AdamW with selected stopping; the matched probe caps training at 1,000 cells per patient and uses a 64-unit MLP for 40 epochs. Other has oracle F1 of 0.886 in the former but 0.497 in the latter (conditional expected priors 0.490/0.487). The recipes also differ in APT preprocessing (Table 4): log1p of raw counts in the matched probe versus provided APT expression without an added log transform in the full-budget MLP, each followed by training-set standardization. Their score differences therefore cannot be attributed to training-cell budget alone. Neither result justifies declaring Other universally identifiable or unidentifiable. The full-budget lineage plot and neural-reference replays are supporting evidence in Appendices B.4–B.5.

5.3 Paired APT Improves on RNA, without Resolved Pairing Specificity

Under the matched probe, MLP fine SB-Macro-F1 increases from 0.589 for RNA to 0.607 for RNA+APT: $+0.0180$ (joint 95% CI $[0.009, 0.030]$). At 5,000 training-selected HVGs, it increases from 0.596 to 0.612: $+0.0165$ ($[0.009, 0.029]$). SGD also improves at both widths (Table 2). The increment is therefore not specific to the 2,000-HVG setting, but remains relative to the tested RNA representations and models.

RNA HVGs	Comparator to paired RNA+APT	SGD Δ [95% CI]	MLP Δ [95% CI]
2,000	RNA	$+0.0185$ $[0.0034, 0.0332]$	$+0.0180$ $[0.0092, 0.0297]$
2,000	RNA + noise	$+0.0394$ $[0.0240, 0.0537]$	$+0.0341$ $[0.0208, 0.0448]$
2,000	Patient shuffle	$+0.0059$ $[-0.0049, 0.0165]$	$+0.0088$ $[0.0014, 0.0145]$
2,000	Patient $\times$ lineage shuffle	$+0.0064$ $[-0.00437, 0.01657]$	$+0.0064$ $[-0.00018, 0.01180]$
5,000	RNA (width sensitivity)	$+0.0217$ $[0.0104, 0.0339]$	$+0.0165$ $[0.0093, 0.0289]$

Table 2: Overall fine SB-Macro-F1 differences, positive favoring paired RNA+APT. Shuffle comparators append shuffled APT to RNA. All 2,000-HVG contrasts use the same targeted experiment: three training seeds and three realizations per seed for each shuffle type. The 5,000-HVG row is the separate matched-width RNA comparison, not a conditional-shuffle test. Scores are computed per fit before averaging; 2,000 paired seed/patient bootstrap replicates additionally resample shuffle realizations where applicable. Intervals are pointwise and exploratory. Both patient-by-lineage-shuffle intervals include zero. Differences are calculated before rounding; displayed marginal scores need not subtract to the displayed difference.

What the controls establish. In the targeted 2,000-HVG experiment, MLP exceeds RNA+noise by 0.0341 and patient-shuffled APT by 0.0088. The latter uses three shuffle draws per training seed, as does the more restrictive patient-by-lineage control. Paired minus patient-by-lineage-shuffled APT is $+0.0064$ ($[-0.00018, 0.01180]$); SGD's corresponding interval also includes zero. This is an unresolved overall contrast, not equivalence or evidence of no pairing information. All control contrasts in the table come from the same targeted experiment; the separate one-shuffle-per-seed width grid is labeled as a sensitivity analysis in the appendix. Lineage-specific contrasts remain exploratory supporting analyses (Appendix B.3).

6 Discussion

APT-Bench separates two empirical questions: the recoverability of cell identity at different hierarchy levels, and the incremental contribution of paired APT beyond a specified RNA reference. The completed MLP oracle and training-prior comparison shows additional discrimination relative to the tested training-only lineage priors. Its heterogeneity across lineages cautions against summarizing APT as uniformly informative or uninformative. The paired-input increment asks a different question. It persists over the two tested RNA widths, but the repeated patient-by-lineage-shuffle contrast remains unresolved overall. Improvements over RNA alone therefore do not establish finer-grained pairing specificity. Patient-disjoint testing and subject-balanced scoring provide the common evaluation foundation, not an additional claim of methodological novelty.

A reusable comparison setting. The resource specifies an APT input, operational ontology, fixed patient folds, metrics, and reference results. Soft-cascade is a hierarchy-aware reference, not a claimed algorithmic advance. These references establish comparison points for improving APT cell typing without changing the input modality, ontology, or held-out patient partitions. Complete data access, permissions, annotation provenance, and a portable evaluator remain necessary release requirements. The completed full-cell-budget MLP adds a non-Transformer reference; more comprehensive APT-compatible annotation baselines and matched-objective comparisons remain useful extensions.

Boundaries of the evidence. The APT cohort contains 40 participants and RNA-derived reference labels. Independent annotation checks and additional APT cohorts are needed to assess broader generalization. Patient-disjoint testing does not remove disease-correlated acquisition effects; this manuscript makes no disease classification or clinical-validation claim. RNA supplies the reference labels, and its apparent difficulty depends on features, preprocessing, and the fitted reference. An APT increment over either 2,000 or 5,000 HVGs may reflect restricted RNA representation; it is not evidence of irreducible information absent from RNA. True-lineage masking and shuffling are privileged diagnostics, and the latter does not identify a causal or molecular mechanism. Three training seeds and pointwise resampling intervals also provide limited uncertainty evidence.

7 Conclusion

APT-Bench establishes a patient-disjoint evaluation setting for cell typing from single-cell aptamer profiles, with a fixed label ontology, explicit subject-balanced scoring, and classical and neural reference results. It asks which hierarchy levels are recoverable in new patients and whether APT contributes additional typing information beyond an RNA reference. Training-prior-controlled oracle decoding reveals usable but heterogeneous within-lineage discrimination under the tested model. The repeated paired probe improves over the tested RNA-only references at both widths, while its overall advantage over patient-by-lineage-shuffled APT remains unresolved. Distinguishing these findings avoids equating improved reference reconstruction with established subtype-specific complementarity.

AI use statement

Generative-AI assistance through ChatGPT and Codex was used in the revision workflow for research-design suggestions, literature search, manuscript editing, analysis-code development, debugging, experiment orchestration, and scientific figure preparation. AI-generated suggestions were not themselves experimental evidence; reported computational results are tied to saved run artifacts. TODO: Authors: confirm the complete tool/model inventory and versions, describe the human checks actually performed on code, citations, figures, and statistical claims, and affirm responsibility for the final submission. Automated checks do not substitute for this author verification.

Ethics statement

APT-Bench is intended for research evaluation, not clinical decision-making. The small single-cohort sample and unresolved acquisition effects preclude claims of clinical validity or demographic fairness. Strong patient-specific signals also create a privacy concern: removing direct identifiers alone does not establish that cell profiles cannot be linked to their source individuals. Any data release must respect participant consent and applicable access terms. TODO: Data owners: supply the actual ethics approval or exemption and consent terms; confirm de-identification procedures, permitted sharing/access controls, and conflicts of interest. These facts have not been verified from the available analysis artifacts; preserve reviewer anonymity when providing documentation.

Reproducibility statement

The repository records the fixed five-fold partition in `benchmark/splits/patient_folds.json`, selected artifact checksums and structural audits in `benchmark/release_audit/`, and artifact-level reproduction commands in `REPRODUCIBILITY.md`. The latter distinguishes rebuilding figures/PDFs from rerunning models. Per-experiment protocol files record sampling settings and seeds. The findings-focused display builder reads completed aggregate results without retraining; its source-to-table mapping and the exact bootstrap audit are recorded in `analysis/FINDINGS_FOCUS_REVISION.md`. A snapshot of the local analysis environment is provided, but is not a lockfile for all historical training runs. TODO: Finalize anonymous reviewer access, data permissions and licenses, archival versioning, original training environments, and a portable raw-data-to-results evaluator. The present repository is not yet a complete independently reproducible benchmark release.

AUTHOR CONTRIBUTIONS

TODO: Add author contributions using CRediT roles (for example: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing, supervision, and funding acquisition). Preserve anonymity in the review version if required.

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TODO: Add funding sources and grant numbers, computational resources, data or clinical contributors, and non-author assistance; verify whether any acknowledgment must be omitted or anonymized during double-blind review.

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A Transformer Reference Implementation Details

This appendix gives the implementation details for the hierarchy-aware reference ladder introduced in Section 4: a flat Transformer, its HCE counterpart, and Soft-cascade Transformer. The cascade focuses on the cell-level hierarchy, lineage $\rightarrow$ subtype. The reference recipes differ in auxiliary supervision as well as hierarchy structure; their comparison does not isolate individual components.

A.1 Cell Encoder

Each cell is represented by a standardized APT vector $x \in \mathbb{R}^{293}$. We embed feature identity and feature value separately. For aptamer index i, we form a token

$$t_i = \mathrm{Emb}(i) + \mathrm{MLP}_{\mathrm{val}}(x_i), \qquad i = 1, \ldots, 293. \tag{2}$$

A learnable `[CLS]` token is prepended to the resulting sequence and processed by a pre-norm Transformer encoder with 4 layers, 8 attention heads, hidden dimension 256, feed-forward dimension 512, GELU activations, and dropout 0.1. The final `[CLS]` representation

$$z = \mathrm{Encode}(x) \in \mathbb{R}^{256} \tag{3}$$

serves as the shared cell representation for downstream tasks.

A.2 Soft-Cascade Reference

Let $M = 5$ be the number of coarse lineages and $K = 27$ the number of fine subtypes. Each subtype k belongs to exactly one parent lineage $m(k)$. A coarse head produces lineage logits

$$c = W_c z \in \mathbb{R}^M, \tag{4}$$

with routing distribution

$$p(m \mid x) = \mathrm{softmax}(c/\tau)_m, \tag{5}$$

where $\tau = 1$.

Instead of a hard cascade, which would force subtype prediction to follow only the top predicted lineage, Soft-cascade Transformer uses a soft mixture of a global subtype path and lineage-specific experts:

$$s_k(x) = \underbrace{g_k(z)}_{\text{global path}} + \alpha\, p(m(k) \mid x)\, e_{m(k),k}(z) + \gamma \log\big(\epsilon + (1-\epsilon) p(m(k) \mid x)\big), \tag{6}$$

Here $g(z)$ is a global subtype head, $e_{m,k}(z)$ is the expert associated with lineage m, $\alpha = 1.0$, $\epsilon = 0.1$, and γ is ramped from 0 to 0.5 during early training. The global path retains a direct subtype-logit contribution alongside the routed expert contribution; we do not isolate its effect from the other components. Final predictions are

$$\hat{k} = \arg\max_k s_k(x), \qquad \hat{m} = \arg\max_m c_m. \tag{7}$$

A.3 Cross-Disease Contrastive Regularization

The reference model includes a cross-disease supervised contrastive term. Let

$$u = \frac{\mathrm{Proj}(z)}{\|\mathrm{Proj}(z)\|_2} \tag{8}$$

be a normalized projection. For anchor i, P_i contains same-subtype, different-disease cells, and N_i contains different-subtype cells. Self-pairs and same-subtype/same-disease pairs are excluded. The implementation uses positive probability mass, rather than the standard average of individual positive log-probabilities:

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{|V|}\sum_{i \in V} \log \frac{\sum_{j \in P_i} \exp(u_i^\top u_j/0.1)}{\sum_{j \in P_i \cup N_i} w_{ij} \exp(u_i^\top u_j/0.1)}, \tag{9}$$

where $V = \{i : |P_i| > 0\}$. Anchors without positives are excluded; a batch with no valid anchor returns zero loss. Although the code permits extra weight on same-lineage negatives, the inspected resolved configuration uses $w_{ij} = 1$ for every included pair. The contrastive multiplier is zero through epoch 5 and then $0.2 \min(1, (t-5)/10)$. This disease-supervised objective is not used by Flat or HCE and is not an independently validated mechanism.

A.4 Optimization and Model Scope

The classification cascade is trained jointly using

$$\mathcal{L} = \lambda_{\text{fine}}\mathcal{L}_{\text{fine}} + \lambda_{\text{coarse}}\mathcal{L}_{\text{coarse}} + w_{\text{contrast}}(t)\mathcal{L}_{\text{contrast}}, \tag{10}$$

where $\mathcal{L}_{\text{fine}}$ and $\mathcal{L}_{\text{coarse}}$ are cross-entropy losses with $\lambda_{\text{fine}} = 1.0$ and $\lambda_{\text{coarse}} = 0.5$. The contrastive weight increases to 0.2 according to the schedule in §A.3.

We optimize the cascade using AdamW with learning rate 10^{-4} and weight decay 10^{-5}, gradient clipping at norm 1.0, batch size 512, and mixed-precision training. Models are trained for at most 50 epochs with early stopping based on validation fine-level macro-F1 with patience 10. The final configuration does not use class-balanced sampling because it underperformed natural sampling after matching the number of optimization steps; the development-only variant was not retained.

Backbone-matched Transformer references. The flat FT-style Transformer reference (Gorishniy et al., 2021) removes the cascade while retaining the identical APT tokenizer, Transformer dimensions, optimizer, early stopping, and training budget above. We call it FT-style because its tokenizer is exactly the APT feature tokenizer used by Soft-cascade Transformer, rather than the original numerical/categorical FT-Transformer tokenizer. This model and its HCE counterpart predict the same $M + K$ lineage-and-subtype nodes. The flat model applies independent cross-entropy to the lineage and subtype logit slices. For HCE (Cultrera di Montesano et al., 2026), let $R_{ij} = 1$ when node j is node i or its descendant, $p = \mathrm{softmax}(q)$ over all nodes, and $s = Rp$. We minimize $-\lambda_{\text{fine}} \log s_{M+k} - \lambda_{\text{coarse}} \log s_m$ for observed subtype k and lineage m. Because every fine label here is a leaf, the fine HCE term alone would reduce to ordinary CE; including the simultaneously observed parent label makes this a stated two-level adaptation rather than an unqualified reproduction of the mixed-granularity setting in the original work. Coarse HCE decoding maximizes subtree scores s_m, not raw internal-node logits; fine decoding maximizes leaf logits. All three Transformer models are evaluated from the checkpoint selected by patient-validation fine macro-F1, without an additional post-selection calibration stage. Table 3 makes the remaining objective differences explicit. In particular, only the soft-cascade reference uses disease labels to construct cross-disease contrastive pairs; Flat and HCE use no disease labels and no contrastive term. The ladder therefore holds backbone capacity, optimization budget, folds, and checkpoint selection, but it is not an objective-matched component ablation. We do not attribute any score difference to soft routing, hierarchical loss, or contrastive regularization in isolation.

Component	Flat FT-style Transformer	FT-style Transformer + HCE	Soft-cascade Transformer
APT tokenizer and Transformer dimensions	Same	Same	Same
Patient folds, optimizer, budget, checkpoint rule	Same	Same	Same
Fine-label supervision	Independent CE	Leaf HCE NLL	CE
Coarse-label supervision	Independent CE	Ancestor HCE NLL	CE
Cross-disease supervised contrastive term	No	No	Yes ($w \leq 0.2$)
Disease labels used by cell model	No	No	Contrastive pairing only
Global subtype path and lineage experts	No	No	Yes
Soft lineage routing	No	No	Yes

Table 3: Components of the three neural references. "Same" denotes an identical specification, not shared fitted weights. The models match the tokenizer, encoder dimensions, data folds, optimizer, maximum epochs, early stopping rule, and checkpoint criterion, but they do not match the complete training objective. In particular, disease labels enter only the soft-cascade contrastive pairing rule. Consequently, comparisons among these rows evaluate complete recipes and are not component ablations.

The main cell-identity results are reported in Table 1; the following sections supply their implementation and diagnostic details.

B DATASET AND EVALUATION DETAILS

B.1 DATASET CONSTRUCTION AND DOCUMENTATION CHECKLIST

The available study workflow establishes the high-level sequence used to construct APT-Bench: clinical peripheral-blood specimens from five cancer groups and healthy controls are converted to single-cell suspensions; cell-profile and viability quality control precedes a paired single-cell transcriptome and 293-aptamer assay; separate transcriptome and aptamer-barcode libraries are linked at the cell level; and transcriptome-derived clusters and annotations provide the cell labels. For a benchmark release, this overview must be accompanied by the following provenance and protocol fields.

Outstanding disease-confounding validation. TODO: Obtain a sample-linked acquisition manifest from the data owner: patient and sample IDs, collection and processing dates, APT assay batch, plate/pool, reagent lot, library-preparation batch, sequencing run, site, and available fresh/frozen and handling records. Audit disease-by-batch overlap before choosing a validation design. Where disease groups have adequate cross-batch support, evaluate patient-disjoint, held-out-batch generalization with all preprocessing and selection restricted to training data. If disease and batch are inseparable, obtain crossed-batch samples or an independent APT cohort; statistical adjustment alone cannot identify their separate contributions. The NC matrices must not be treated as external validation if they are the healthy subset of this cohort. These validations are pending, not completed. Expression and QC matrices alone do not establish acquisition provenance; archived disease analyses cannot establish disease-specific biological effects.

Transcriptomic preprocessing and cell quality control. The transcriptomic count matrix was processed with Seurat following its standard workflow of normalization, feature selection, dimensionality reduction, neighborhood-graph construction, and clustering (Hao et al., 2021). DoubletFinder was used to identify and remove predicted doublets (McGinnis et al., 2019). Low-quality cells were additionally filtered using per-cell UMI count, detected-gene count, and mitochondrial-transcript fraction. These steps preceded cluster annotation and construction of the linked APT benchmark cohort. TODO: Report the Seurat and DoubletFinder versions, every parameter and random seed, the exact UMI/gene/mitochondrial thresholds, and cell counts before and after each filter.

Manual annotation and reference-label reliability. After Seurat clustering, investigators manually assigned lineage and subtype names by inspecting canonical marker-gene expression reported in prior PBMC and pan-cancer studies (Zhang et al., 2024; Yang et al., 2024). No quantitative annotation confidence, inter-annotator agreement, or external reference-transfer score is available. Accordingly, these labels are operational reference annotations, not error-free biological ground truth, and benchmark scores are conditional on this manual taxonomy. The *Unknown* label was assigned when a cluster did not show a coherent canonical marker pattern for the expected PBMC subtypes; this was a manual reject decision rather than a calibrated confidence threshold. Unknown cells are excluded from the primary 27-subtype benchmark. TODO: Provide the full subtype-by-marker citation table, the number and expertise of annotators, whether they were blinded to disease, the adjudication procedure, and the explicit rule used to assign Unknown.

Cross-disease semantics and hierarchy mapping. The same 27 subtype labels and definitions were applied across all six disease groups; no subtype was renamed or redefined for a particular disease. This gives the labels a consistent operational meaning, but it does not prove that marker expression or biological state is invariant across diseases. Each subtype is assigned to one of five parent lineages by a manually curated lookup. The exported lookup and structural audit are available under `benchmark/release_audit/`: the existing subtype summary has 27 unique subtypes, five parents, and 361,792 cells, and the fold manifest has five disjoint groups of eight patients. These checks establish internal bookkeeping consistency, not independent biological validity. TODO: Document independent validation of the subtype-to-lineage mapping, including reviewers, disagreements, adjudication, and any marker-based checks.

- TODO: Report recruiting site(s) and dates, prospective versus retrospective collection, exact patient count per group, inclusion/exclusion and diagnostic criteria, and whether each participant contributed exactly one specimen.
- TODO: Report available demographics and clinical covariates, including age, sex, disease stage, treatment status, and comorbidities; summarize missingness and state which fields cannot be released.
- TODO: Report blood volume, collection tubes, time to processing, cell-isolation and cryopreservation procedures, quality-control thresholds, and sample/cell exclusion counts.
- TODO: Report assay/platform and reagent versions, sequencing instruments and depth, 293-probe panel design, and a verified protein-target manifest or explicitly state why only anonymized probe identifiers can be released.
- TODO: Report barcode-matching rates, APT normalization and filtering, all remaining software versions and parameters, and whether annotation was performed before investigators examined APT profiles or disease labels.
- The documented final cell-flow accounting is 374,854 linked post-QC cells − 13,062 Unknown cells = 361,792 benchmark cells. TODO: Data owners: provide participant and cell counts at all earlier collection/QC exclusions before a complete cohort-flow diagram can be drawn. Supply the approval, consent, access, and licensing evidence listed in the Ethics and Reproducibility Statements.

B.2 Benchmark Interface and Release Contract

The cell-typing interface records pseudonymous cell and patient IDs, fixed folds, feature IDs, reference and predicted labels, and ontology columns. A complete release requires the processed paired matrices, permissions and access procedure, reference configurations, and a raw-data-to-evaluator command. The current split manifest, selected artifact checksums and rebuild commands cover only part of that release. TODO: Finalize anonymous data/code access, licenses, original training environment locks, archival versioning, and the portable end-to-end command. Where access is controlled, provide the complete schema and access procedure.

B.3 Hierarchy and Conditional-Pairing Protocol

Design and training conditions. This analysis is exploratory: its design followed inspection of a single-seed pilot. At both RNA widths, log-count normalization and mean-binned dispersion selection use only the current inner or outer training partition. All 361,792 cells are barcode-aligned. RNA is an internal reconstruction reference because it also supplies the operational labels.

Table 4 specifies both training recipes. For the matched probe, weights use only the current training subset. Writing N for its cell count, $n_{p(i)}$ for the count from the patient of cell i, and n_{y_i} for its class count, the implemented weights are

$$a_i = \frac{1}{n_{p(i)} n_{y_i}}, \qquad w_i = \frac{a_i}{N^{-1} \sum_{j \in \text{train}} a_j}.$$

This is the mean-normalized product of inverse patient frequency and balanced-class weights. The product does not ensure equal total weight per patient or per class. The full-budget MLP instead uses unweighted CE.

Setting	Matched SGD	Matched MLP	Full-budget MLP
Model / loss	One-vs-rest logistic; weighted log loss	64-unit ReLU; weighted cross-entropy	256/128-unit ReLU; dropout 0.1; unweighted cross-entropy
Training cells	Same fixed subset, at most 1,000 cells per training patient		All available training cells
Optimizer / rate	Averaged SGD; optimal schedule $\eta_t = 1/[\alpha(t + t_0)]$	Adam; initial rate 10^{-3}	AdamW; rate 10^{-3}
Update size / epochs	One cell per update; 40 passes; no tolerance stopping	Batch 512; 40 epochs; no early stopping	Batch 1,024; select epoch 1–50 on inner validation
Regularization grid	L2 $\alpha \in \{10^{-4}, 10^{-3}\}$	L2 $\alpha \in \{10^{-3}, 10^{-1}\}$	Weight decay $\{10^{-5}, 10^{-3}\}$
APT input	$\log(1 +$ raw APT count$)$, then featurewise training mean/SD standardization		Provided `apt_expression`, then training mean/SD standardization; no added log transform
RNA input	Per-cell counts scaled to 10^4, log1p; training-selected 2,000/5,000 HVGs; training SD scaling without centering		Not used
Noise control	293 independent $\mathcal{N}(0, 1)$ features appended to scaled RNA; no further noise scaling; separate train/test draws		Not used

Table 4: Core experiment configurations read from the implemented recipes. The SGD offset t_0 follows the library's optimal-schedule heuristic. Regularization parameters have estimator-specific meanings, not a shared penalty scale. Each outer fold uses the next frozen patient fold for inner validation; highest validation SB-Macro-F1 selects the grid value (and full-budget epoch), retaining the first candidate/earliest epoch on ties. Refits reinitialize on the outer development partition. Transform statistics, HVG selection and sample weights use only the current training partition; per-cell RNA library normalization does not estimate population parameters. The upstream transform underlying provided APT expression remains a provenance requirement, not an assumed raw-count equivalence.

Oracle and prior controls. Oracle decoding masks frozen fine scores to children of the true parent without changing routing or retaining only correctly predicted parents. Conditional majority and proportion priors use the actual training subset's subtype frequencies within that parent. The proportion prior contributes its expected confusion matrix, which is then scored; this is not expected finite-sample F1. These controls are privileged-label diagnostics, not deployable predictions. For an individual lineage, restrict true cells to that lineage but retain all prediction columns, so predictions outside the lineage remain false negatives. Normalize each eligible patient's within-lineage matrix, pool and score its fixed child classes. Patients with zero scope support are excluded. This differs from Table 1, where normalization uses each patient's entire evaluated cell population.

Conditional shuffling. The targeted 2,000-HVG experiment uses three shuffle realizations per training seed for each control: within patient and within patient and true lineage. Entire APT vectors are permuted separately in training and test partitions, preserving each group's empirical APT distribution. The latter control uses privileged true test lineage. Singleton and unchanged-pair fractions are retained in the reproducibility outputs. Each realization has its own selected/refitted control model; it is not just a test-time perturbation of the paired model.

Statistical target and exact resampling order. Let $C^{(p)}_{s,m}$ denote the confusion matrix for training seed s, condition m and patient p in the chosen scope, with mass n_p. For patient multiplicities $w = (w_1, \ldots, w_{40})$, define

$$F_{s,m}(w) = \mathrm{MacroF1}_{\mathcal{K}}\left(\sum_{p:n_p>0} w_p C^{(p)}_{s,m}/n_p\right),$$

where $\mathcal{K}$ is the full ontology or the fixed child classes for a lineage. Dividing this pooled matrix by its total weight would not change F1. There is no pooling of confusion matrices across training seeds or shuffle realizations before computing F1. The paired-versus-shuffled point estimate is

$$\widehat{\Delta} = \frac{1}{3}\sum_{s=1}^{3}\left[F_{s,\mathrm{paired}}(\mathbf{1}) - \frac{1}{3}\sum_{r=1}^{3} F_{s,\mathrm{shuffle}(r)}(\mathbf{1})\right].$$

For each of 2,000 bootstrap replicates:

1. Draw 40 patients with replacement from the 40 held-out patients, giving one multinomial multiplicity vector w shared across conditions, seeds and shuffle realizations in the paired comparison.
2. Draw three seed indices s_1, s_2, s_3 with replacement from the three observed training seeds; a replicate does not select just one seed.
3. For each selected seed slot j, draw three shuffle indices with replacement from its three realizations. Compute each F1 separately on the shared patient draw, average these three control F1 values, and subtract that average from $F_{s_j,\mathrm{paired}}(w)$.
4. Average the three seed-slot differences. Report the 2.5th and 97.5th percentiles of finite replicates. A scope with no eligible resampled patient is undefined, not assigned zero.

For non-shuffled comparisons the control has one value per seed, so step 3 does not resample a control realization. Full-budget MLP and RNA-width intervals likewise average seed-specific F1 values after paired patient resampling. Frozen-reference intervals in Table 1 resample patients only. These pointwise intervals do not correct for multiple comparisons, repeat model selection, or account for all analysis adaptation; three seeds provide limited training-uncertainty evidence.

RNA-width sensitivity. The 2,000/5,000-HVG reference grids use three seeds but one patient-shuffle realization per seed; their point estimates are not substitutes for the three-shuffle targeted contrast in Table 2. The 5,000-HVG paired-versus-RNA result is retained regardless of outcome. The more restrictive conditional shuffle was evaluated only at 2,000 HVGs; its unresolved overall interval was not followed by additional training to seek significance. Full run schedules and the pre-existing follow-up selection rule are documented in the reproducibility protocol.

Model	Scope	APT	APT oracle	RNA oracle	Prior MAP	Prior expected CM
SGD	All	0.089	0.289	0.542	0.148	0.182
SGD	B	0.133	0.334	0.556	0.176	0.199
SGD	Myeloid	0.147	0.201	0.661	0.072	0.124
SGD	NK	0.220	0.840	0.791	0.434	0.494
SGD	Other	0.018	0.470	0.603	0.496	0.487
SGD	T	0.119	0.184	0.379	0.067	0.098
MLP	All	0.093	0.309	0.672	0.148	0.182
MLP	B	0.212	0.405	0.750	0.176	0.199
MLP	Myeloid	0.162	0.211	0.774	0.072	0.124
MLP	NK	0.241	0.849	0.839	0.434	0.494
MLP	Other	0.076	0.497	0.652	0.496	0.487
MLP	T	0.119	0.191	0.535	0.067	0.098

Table 5: Matched-budget hierarchy diagnostics: mean over three training seeds, 2,000 training-selected RNA HVGs, identical capped training subsets and patient folds. All-cell rows normalize each patient's complete matrix; lineage rows instead normalize within that lineage and score fixed children. Their scores cannot be averaged to recover all-cell SB-Macro-F1. Oracle and prior controls receive true lineage. Expected-CM F1 is not expected finite-sample F1. Other has 38 eligible patients; other scopes have 40. These are descriptive point estimates, not paired APT–RNA superiority tests.

HVGs	Task	Model	RNA	Paired	Noise	Patient shuffle	Paired minus RNA
2,000	Coarse	SGD	0.745	0.751	0.745	0.750	+0.0063 [0.0004, 0.0127]
2,000	Coarse	MLP	0.871	0.872	0.868	0.874	+0.0008 [-0.0057, 0.0075]
2,000	Fine	SGD	0.432	0.450	0.411	0.442	+0.0185 [0.0033, 0.0332]
2,000	Fine	MLP	0.589	0.607	0.573	0.597	+0.0180 [0.0092, 0.0298]
5,000	Coarse	SGD	0.772	0.777	0.773	0.777	+0.0047 [0.0016, 0.0079]
5,000	Coarse	MLP	0.873	0.881	0.876	0.878	+0.0086 [0.0036, 0.0144]
5,000	Fine	SGD	0.456	0.478	0.437	0.468	+0.0217 [0.0104, 0.0339]
5,000	Fine	MLP	0.596	0.612	0.586	0.606	+0.0165 [0.0093, 0.0289]

Table 6: RNA-width reference sensitivity: three training seeds and five patient folds. Point estimates average seed-specific SB-Macro-F1; paired differences have joint seed/patient 95% intervals. Patient shuffle uses one draw per seed in this grid; the targeted 2,000-HVG experiment uses three draws per seed for each shuffle type. All normalization and HVG selection are training-partition-only. Noise adds 293 dimensions. Differences are computed before rounding; differences and interval endpoints use four decimal places.

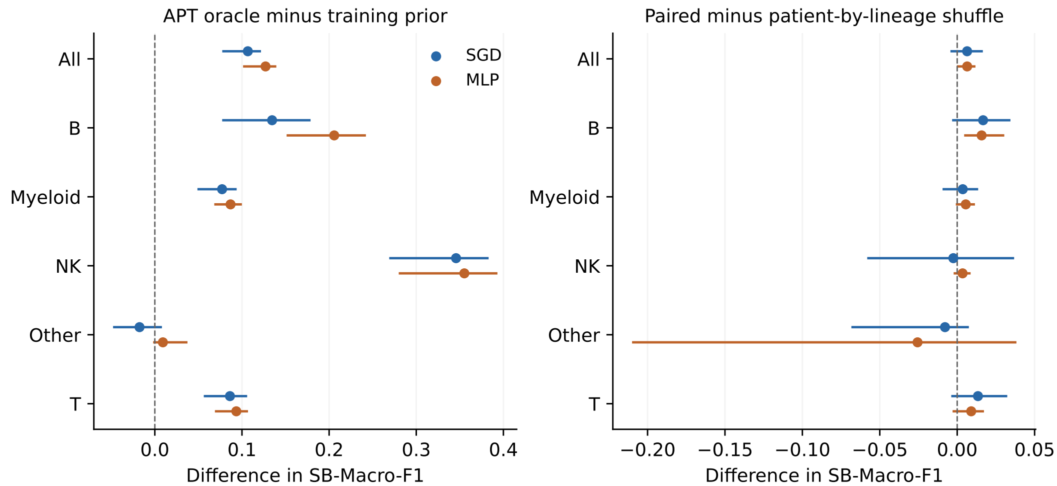

Figure 3: Exploratory scope-specific contrasts at the matched 2,000-HVG budget. Left: APT oracle minus the conditional-proportion expected-CM prior. Right: paired RNA+APT minus patient-by-lineage-shuffled APT. Pointwise intervals follow the joint resampling specified above; all lineages are retained. All-cell and within-lineage scores have different patient-normalization populations. Axes differ across panels. Overall right-panel intervals include zero; subgroup contrasts do not override that result.

B.4 Full-Budget Hierarchy Support

The full-budget 256/128-unit MLP uses all development cells and a different optimization recipe from the capped 64-unit matched-input MLP. Its overall ordinary/oracle/expected-prior scores are 0.141/0.385/0.180. The oracle-minus-prior difference is $+0.205$ ($[0.174, 0.217]$). Figure 4 reports every lineage. Differences from the matched-input experiment cannot be attributed to training-cell budget alone, because APT preprocessing, architecture, optimization and selection also differ (Table 4).

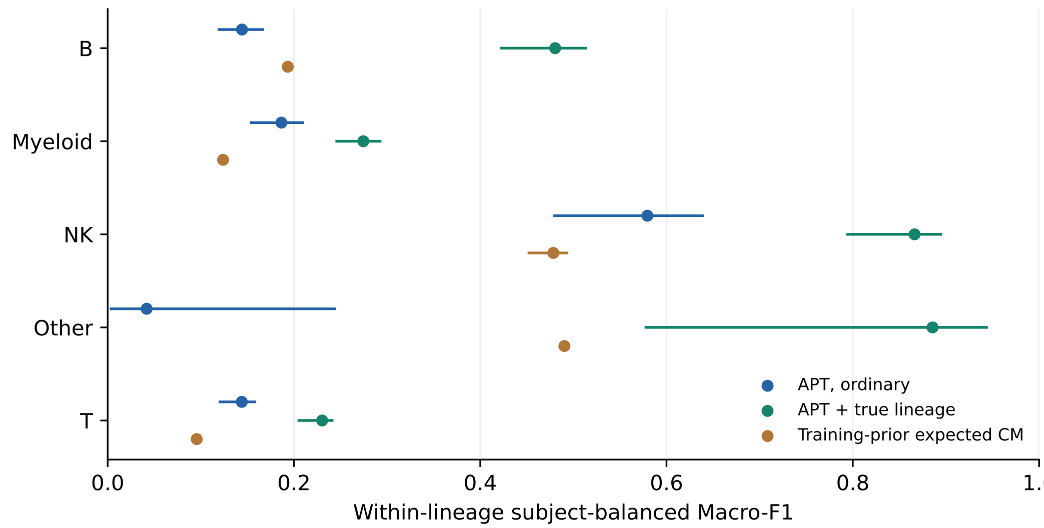

Figure 4: Full-budget MLP support: three-seed means and pointwise joint seed/patient 95% intervals. Each eligible patient's matrix is normalized within the displayed lineage, then pooled before Macro-F1 is computed over its fixed children. Outside-lineage predictions remain false negatives. These scores cannot be averaged across lineages to reconstruct Table 1. The conditional-proportion baseline is F1 of an expected confusion matrix, not expected F1. Oracle information is privileged. This is not the capped-budget APT–RNA comparison in Figure 2.

B.5 True-Lineage Restricted Decoding

We replay the 15 frozen final checkpoints (three neural references, five patient-disjoint folds), retain explicit cell IDs and ontology columns, and require exact reconstruction of their original coarse and fine predictions before computing this diagnostic. For each of the same 361,792 held-out cells, we mask the fine score vector to subtypes whose parent is the true lineage, then take its argmax. We do not change Soft-cascade routing or condition the evaluation set on a correct coarse prediction. The diagnostic therefore uses privileged test-label information and cannot be placed on the deployable leaderboard. Table 7 reports subject-balanced pooled Macro-F1 and paired patient-bootstrap differences, conditional on these frozen models. Reduced candidate-set size contributes to the gain; neither the gain nor the remaining error identifies an assay-specific information ceiling.

Model	Ordinary fine	True-lineage oracle	Paired gain [95% CI]
Flat Transformer	0.139	0.366	+0.227 [0.202, 0.238]
HCE Transformer	0.143	0.361	+0.218 [0.196, 0.231]
Soft-cascade Transformer	0.145	0.358	+0.214 [0.194, 0.230]

Table 7: Privileged true-lineage restricted decoding on the same frozen predictions. Scores are subject-balanced pooled Macro-F1; gains use 2,000 paired patient-bootstrap resamples and pointwise intervals, conditional on fitted models. The oracle knows the true parent and is not a deployable method.

B.6 Complete Per-Subtype Reporting

Table 8 retains all 27 subtypes, cell counts, patient coverage, and historical cell-weighted per-class F1 with 2,000 patient-bootstrap intervals. These are not the subject-balanced main metric. They support transparent class-level inspection, not a biological interpretation of marginal support–performance correlations.

Table 8: Complete subtype support and cell-weighted per-class OOF F1 (not SB-F1), with pointwise 95% patient-bootstrap intervals. Counts and patient coverage refer to the same 361,792 cells.

Subtype	Lineage	Cells	Patients	Soft-cascade	XGBoost
Erythrocyte	Other	65	6	0.253 [0.000, 0.395]	0.133 [0.000, 0.545]
Atypical Memory B	B	781	34	0.005 [0.000, 0.014]	0.005 [0.000, 0.014]
CD4 Monocyte-4	Myeloid	1,405	35	0.027 [0.006, 0.053]	0.014 [0.000, 0.033]
platelet	Other	1,574	38	0.078 [0.016, 0.125]	0.063 [0.015, 0.106]
Tcm	T	2,121	36	0.000 [0.000, 0.000]	0.004 [0.000, 0.016]
Memory B-2	B	2,130	33	0.017 [0.004, 0.033]	0.011 [0.000, 0.024]
Plasma	B	2,173	40	0.611 [0.522, 0.684]	0.599 [0.496, 0.685]
pDC	Myeloid	2,615	40	0.029 [0.010, 0.061]	0.019 [0.005, 0.040]
CD14 Monocyte-3	Myeloid	2,977	32	0.258 [0.170, 0.306]	0.177 [0.103, 0.211]
CD8 + CTL-3	T	3,244	27	0.020 [0.004, 0.035]	0.023 [0.008, 0.043]
GCB	B	3,283	31	0.039 [0.007, 0.071]	0.047 [0.001, 0.073]
Cycling T	T	3,667	40	0.028 [0.015, 0.043]	0.012 [0.003, 0.022]
CD14 Monocyte-2	Myeloid	3,931	40	0.075 [0.046, 0.102]	0.078 [0.056, 0.097]
CD8 + CTL-2	T	4,049	33	0.069 [0.017, 0.116]	0.078 [0.015, 0.113]
CD16 Monocyte-2	Myeloid	5,142	40	0.111 [0.064, 0.161]	0.117 [0.063, 0.164]
CD8 + CTL-1	T	7,303	40	0.009 [0.000, 0.024]	0.002 [0.001, 0.004]
cDC2	Myeloid	7,376	40	0.132 [0.099, 0.181]	0.125 [0.079, 0.167]
CD8 Tem-2	T	10,548	40	0.019 [0.005, 0.038]	0.017 [0.008, 0.030]
CD16 Monocyte-1	Myeloid	14,678	40	0.152 [0.111, 0.198]	0.118 [0.090, 0.150]
CD4 + Tcm-2	T	16,861	39	0.188 [0.111, 0.236]	0.188 [0.136, 0.220]
Naive T	T	17,960	40	0.131 [0.067, 0.183]	0.149 [0.073, 0.199]
Memory B-1	B	23,722	40	0.034 [0.020, 0.048]	0.032 [0.017, 0.047]
CD8 Tem-1	T	26,714	40	0.187 [0.128, 0.250]	0.162 [0.107, 0.221]
CD14 Monocyte-1	Myeloid	29,871	40	0.437 [0.378, 0.495]	0.451 [0.392, 0.507]
CD56 NK	NK	38,442	40	0.398 [0.320, 0.436]	0.437 [0.373, 0.480]
CD16 NK	NK	58,507	40	0.383 [0.326, 0.434]	0.376 [0.317, 0.429]
CD4 + Tcm-1	T	70,653	40	0.417 [0.368, 0.460]	0.435 [0.393, 0.473]

B.7 Scoring Units and Supporting Split Comparison

Metric definitions. Let $C^{(p)} \in \mathbb{R}^{K\times K}$ be the confusion matrix for held-out subject p and $n_p = \sum_{ij} C^{(p)}_{ij}$. We distinguish

$$C^{\mathrm{CW}} = \sum_p C^{(p)}, \qquad F_1^{\mathrm{CW}} = \mathrm{MacroF1}(C^{\mathrm{CW}}), \tag{11}$$

$$C^{\mathrm{SB}} = \frac{1}{P}\sum_p \frac{C^{(p)}}{n_p}, \qquad F_1^{\mathrm{SB}} = \mathrm{MacroF1}(C^{\mathrm{SB}}), \tag{12}$$

$$F_1^{\mathrm{within}} = \frac{1}{P}\sum_p \mathrm{MacroF1}(C^{(p)}). \tag{13}$$

We call these *cell-weighted pooled Macro-F1*, *subject-balanced pooled Macro-F1*, and *mean within-subject Macro-F1*. Subject balancing normalizes each subject confusion matrix to unit mass before pooling; it is neither equal-cell subsampling nor averaging per-subject Macro-F1. Released files retain the legacy column name `patient_balanced_macro_f1` for F_1^{SB}, but no different patient-balanced statistic is used.

The numerical split comparison appears in this appendix (Table 9). Its historical scores are cell-weighted in both columns, rather than the subject-balanced leaderboard metric.

Exact subject-normalized pooling is primary. The older equal-cell Monte Carlo sensitivity remains archived, rather than occupying a separate table.

The historical patient-disjoint XGBoost run uses 32 training and eight test patients per fold, without a validation partition; Soft-cascade uses 24/8/8 training/validation/test patients. Both cell-split runs instead use 253,254/54,269/54,269 cells, with all 40 patients represented in each partition. An audit of the two saved cell-level split manifests verified pairwise-disjoint training, validation and test cell-ID sets. Patient-disjoint scores pool 361,792 OOF cells; cell-split scores use 54,269 cells, not a common test set across protocols. The historical Soft-cascade CSV exports used in this split audit, both patient-disjoint pooled OOF and cell-split test predictions, retain patient and label fields but not cell IDs. This limits independent barcode linkage for those exports, not the manifest-based partition check, and does not imply actual cell overlap. The separate frozen-checkpoint oracle replay in Appendix B.5 retains explicit cell IDs; XGBoost prediction IDs were checked against the historical manifests. The split-audit reproducibility record reports full counts, source hashes and historical training-record limitations. This is a descriptive comparison, not an estimate of a pure leakage effect.

Task	**Model**	**Patient-disjoint**	**Cell split**	**Observed difference**
Coarse lineage	XGBoost	0.322	0.372	+0.050
Coarse lineage	Soft-cascade Transformer	0.326	0.417	+0.090
Fine subtype	XGBoost	0.143	0.212	+0.069
Fine subtype	Soft-cascade Transformer	0.152	0.260	+0.108

Table 9: Historical descriptive protocol comparison. Both score columns are cell-weighted pooled Macro-F1 (CW), not the subject-balanced metric of Table 1. Observed difference is cell split minus patient-disjoint, computed from unrounded saved predictions before display rounding. Training, validation and test allocations were not matched; the patient-disjoint column pools five outer folds, whereas cell split evaluates one subset of cells from already-seen patients. Protocol-difference uncertainty was not estimated. These related model/resolution contrasts are not independent replications or causal estimates of leakage.

B.8 Additional Reference Inventory

Table 10 supplies the three additional classical references; Table 1 contains the primary models.

Model	CI	Coarse SB-Macro-F1 ↑	Fine SB-Macro-F1 ↑
Linear SVM	P	0.309 [0.292, 0.324]	0.105 [0.089, 0.118]
Random Forest	P	0.253 [0.227, 0.277]	0.089 [0.079, 0.097]
Reference Correlation	P	0.171 [0.147, 0.192]	0.068 [0.050, 0.078]

Table 10: Additional APT-only references, complementing Table 1. P denotes pointwise 95% patient-bootstrap intervals conditional on frozen fits.

B.9 Archived Analyses

The version archive retains the earlier resource-comparison table, single-seed paired-input pilot, chance adjustment, disease-classification audits, subject–cell allocation and assay-budget analyses, marginal support correlations, and redundant cell-weighted metrics. Composition recovery (including direct ridge and the unresolved lineage result), the incomplete LODO stress test, and RNA-difficulty/rescue/damage analyses are archived in full rather than selectively reported. These analyses do not support the two core cell-identity findings reported here. The revision index identifies the source versions and retained outputs.